Database Management System (DBMS) Quick Notes

Introduction to DBMS

DBMS is software used to store, manage, and retrieve data efficiently. Advantages include reduced redundancy, data consistency, security, and backup management.

Database Models

Hierarchical, Network, Relational, and Object-Oriented models are common database models. The Relational Model is the most widely used.

Keys in DBMS

Primary Key uniquely identifies records. Foreign Key creates relationships between tables. Candidate Key can uniquely identify tuples. Super Key is a set of attributes identifying tuples.

Normalization

Normalization removes redundancy and improves data integrity. Forms include 1NF, 2NF, 3NF, and BCNF.

SQL Basics

DDL commands: CREATE, ALTER, DROP. DML commands: INSERT, UPDATE, DELETE. DQL command: SELECT. TCL commands: COMMIT and ROLLBACK.

Joins

INNER JOIN returns matching rows. LEFT JOIN returns all left table rows. RIGHT JOIN returns all right table rows. FULL JOIN returns all records from both tables.

Transactions

A transaction is a logical unit of work. ACID properties are Atomicity, Consistency, Isolation, and Durability.

Concurrency Control

Used to manage simultaneous transactions. Lock-based protocols and timestamp protocols help avoid conflicts.

Deadlock

Deadlock occurs when transactions wait indefinitely for resources. Prevention, avoidance, and detection techniques are used.

Indexing

Indexes improve search speed. Primary index, secondary index, clustered and non-clustered indexes are commonly used.

Important SQL Commands

Command Type	Examples
DDL	CREATE, ALTER, DROP
DML	INSERT, UPDATE, DELETE
DQL	SELECT
TCL	COMMIT, ROLLBACK